DAY 12: INFRASTRUCTURE AS CODE (IaC) USING TERRAFORM

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1. AIM

In this task, my aim was to understand Infrastructure as Code (IaC) using Terraform and create a local resource using code instead of manual setup.

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2. SYSTEM DETAILS

- Operating System: Windows 10
- Tool Used: Terraform CLI
- Code Editor: VS Code
- Terminal: PowerShell

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3. FILES I CREATED

For this task, I worked with the following file:

- main.tf – Terraform configuration file

After running Terraform, additional files were generated:

- day12_check.txt – created by Terraform
- terraform.tfstate – state file created automatically

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4. STEP BY STEP IMPLEMENTATION

- **STEP 1: INSTALL TERRAFORM**

  I installed Terraform CLI from the official website.

- **STEP 2: CHECK VERSION**

```
terraform -version
```

  This command checks whether Terraform is installed correctly.

  MY UNDERSTANDING

  This command confirms that Terraform is ready to use.

- **STEP 3: CREATE MAIN FILE**

  I created a file named main.tf and added the following code:

```
terraform {
  required_providers {
```

```
    local = {
      source  = "hashicorp/local"
      version = "2.4.0"
    }
  }
}

resource "local_file" "devops_log" {
  filename = "${path.module}/day12_check.txt"
  content  = "Terraform managed this file on April 17, 2026."
}
```

MY UNDERSTANDING

- terraform block → defines provider
- resource → defines what to create
- local_file → creates a file locally
- filename → file name and path
- content → text inside file

🔹 STEP 4: INITIALIZE TERRAFORM

```
terraform init
```

MY UNDERSTANDING

This command downloads required plugins (providers) needed for Terraform to work.

🔹 STEP 5: PLAN INFRASTRUCTURE

```
terraform plan
```

MY UNDERSTANDING

This command shows what Terraform will create before actually creating it.

🔹 STEP 6: APPLY INFRASTRUCTURE

```
terraform apply
```

Then I typed:

```
yes
```

MY UNDERSTANDING

This command actually creates the resource defined in the file.

🔹 STEP 7: VERIFY OUTPUT

After running apply, I checked my folder and found:

- day12_check.txt

This file was created automatically by Terraform.

🔹 **STEP 8: CHECK STATE FILE**

I noticed a file:

- terraform.tfstate

MY UNDERSTANDING

This file stores the current state of infrastructure created by Terraform.

It helps Terraform track what is already created.

🔹 **STEP 9: MODIFY FILE & RE-APPLY**

I updated the content in main.tf:

```
content = "Updated via IaC."
```

Then I ran:

```
terraform apply
```

MY UNDERSTANDING

Terraform detected changes and updated the file automatically.

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5. PROBLEMS I FACED

❌ I didn't understand Terraform state file
✓ Later I understood it stores infrastructure history

❌ I was confused about plan vs apply
✓ After using both, I understood difference clearly

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6. WHAT I LEARNED

- Infrastructure can be written using code
- Terraform automates resource creation
- plan shows preview of changes
- apply executes changes
- state file tracks infrastructure

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7. CONCLUSION

In this task, I successfully created infrastructure using Terraform. I learned how infrastructure can be managed using code instead of manual configuration.

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8. FINAL REFLECTION

Earlier, I thought infrastructure setup is manual work. But now I understand that with IaC tools like Terraform, everything can be automated, version-controlled, and reused easily.